

SQL Homework

Oracle HR Schema

20 Query Writing Tasks

Prepared for DTank54 Group Students

Database: Oracle
Schema: HR (Human Resources)
Level: A1 (Beginner)

Instructions

This is your homework. There are 20 tasks. For each task, you must write a SQL query. Read the question carefully. Think about which table and columns you need. Then write your answer in the space provided.

Rules:

1. Write your SQL queries by yourself. Do not copy from the lesson book.
2. Use the Oracle HR schema tables: REGIONS, COUNTRIES, LOCATIONS, DEPARTMENTS, JOBS, EMPLOYEES, JOB_HISTORY.
3. Use double quotes for column aliases with spaces. For example: AS "Full Name".
4. Use single quotes for text values. For example: 'IT_PROG'.
5. Do not worry if your answer is different from your friend's answer. There are many correct ways to write the same query.
6. Try to finish all 20 tasks. If you can not do a task, skip it and come back later.

Quick Reference - Tables:

Table	Important Columns
REGIONS	REGION_ID, REGION_NAME
COUNTRIES	COUNTRY_ID, COUNTRY_NAME, REGION_ID
LOCATIONS	LOCATION_ID, CITY, COUNTRY_ID
DEPARTMENTS	DEPARTMENT_ID, DEPARTMENT_NAME, MANAGER_ID, LOCATION_ID
JOBS	JOB_ID, JOB_TITLE, MIN_SALARY, MAX_SALARY
EMPLOYEES	EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, JOB_ID, SALARY, COMMISSION_PCT, MANAGER_ID, DEPARTMENT_ID, HIRE_DATE
JOB_HISTORY	EMPLOYEE_ID, START_DATE, END_DATE, JOB_ID, DEPARTMENT_ID

Oracle HR Schema Tables

Tasks

Task 1. Show all columns and all rows from the JOBS table.

Hint: Use SELECT * with the correct table name.

Task 2. Show FIRST_NAME, LAST_NAME, and SALARY from the EMPLOYEES table.

Hint: Write three column names after SELECT.

Task 3. Show all different (unique) JOB_ID values from the EMPLOYEES table.

Hint: Use the DISTINCT keyword.

Task 4. Show all different CITY names from the LOCATIONS table.

Hint: Use DISTINCT with the CITY column.

Task 5. Show FIRST_NAME as "First Name" and LAST_NAME as "Last Name" from EMPLOYEES.

Hint: Use AS with double quotes for aliases with spaces.

Task 6. Show all employees whose SALARY is greater than 8000. Show FIRST_NAME, LAST_NAME, and SALARY.

Hint: Use WHERE with the greater than (>) operator.

Task 7. Show all employees who work in department 80 (DEPARTMENT_ID = 80). Show FIRST_NAME, LAST_NAME, and DEPARTMENT_ID.

Hint: Use WHERE with the equal (=) operator.

Task 8. Show FIRST_NAME, LAST_NAME, and SALARY for employees whose JOB_ID is 'SA_REP' AND whose SALARY is greater than or equal to 10000.

Hint: Use WHERE with AND to combine two conditions.

Task 9. Show all employees whose DEPARTMENT_ID is 10 or 20 or 30. Show FIRST_NAME, LAST_NAME, and DEPARTMENT_ID.

Hint: Use WHERE with the IN operator.

Task 10. Show all employees whose SALARY is between 5000 and 12000. Show FIRST_NAME, LAST_NAME, and SALARY.

Hint: Use WHERE with the BETWEEN operator.

Task 11. Show all employees whose FIRST_NAME starts with the letter 'S'. Show FIRST_NAME and LAST_NAME.

Hint: Use WHERE with LIKE and the % wildcard.

Task 12. Show all employees whose LAST_NAME contains the letters 'an' (for example: Zan, Nancy, etc.). Show FIRST_NAME and LAST_NAME.

Hint: Use LIKE with '%an%' to find the text anywhere in the name.

Task 13. Show all employees who do NOT have a commission (COMMISSION_PCT is NULL). Show FIRST_NAME, LAST_NAME, and COMMISSION_PCT.

Hint: Use IS NULL to check for empty values.

Task 14. Show all employees who HAVE a commission (COMMISSION_PCT is not NULL). Show FIRST_NAME, LAST_NAME, and COMMISSION_PCT.

Hint: Use IS NOT NULL.

Task 15. Show all employees ordered by SALARY from highest to lowest. Show FIRST_NAME, LAST_NAME, and SALARY.

Hint: Use ORDER BY with DESC.

Task 16. Show all employees ordered by FIRST_NAME from A to Z. Show FIRST_NAME and LAST_NAME.

Hint: Use ORDER BY with ASC (or just ORDER BY, because ASC is default).

Task 17. Show FIRST_NAME, LAST_NAME, and SALARY * 12 as "Annual Salary" from EMPLOYEES for department 90 only. Order the result by SALARY from high to low.

Hint: Combine WHERE, arithmetic (*), alias with double quotes, and ORDER BY.

Task 18. Show FIRST_NAME, LAST_NAME, and JOB_ID for employees whose JOB_ID is NOT 'IT_PROG'. Use the != operator or the <> operator.

Hint: Use WHERE with != or <> for "not equal".

Task 19. Show FIRST_NAME, LAST_NAME, and EMAIL joined together like this: 'Steven King (SKING)'. Use the || operator. Give this column the alias 'Employee Info' (double quotes).

Hint: Use || to join columns and text. For example: first_name || ' ' || last_name.

Task 20. Show the current date using SYSDATE from the DUAL table. Give the result column the alias "Today" (double quotes).

Hint: SELECT SYSDATE AS "Today" FROM dual;

Good luck, DTank54!

Remember: Practice makes perfect!